



Automated Wildlife Classification in National Parks

Deep Learning, University of Toronto SCS (3546) - Summer 2025

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The Challenge

Live Competition: Conser-vision: Image Classification



The Problem:

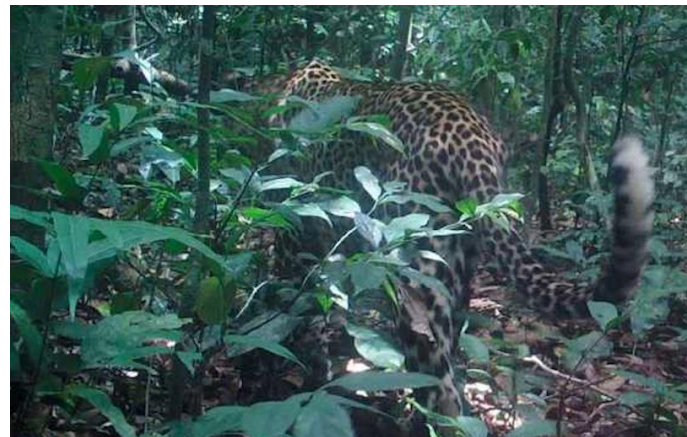
- Manual classification of millions of camera trap images is time-consuming and resource-intensive for wildlife researchers
- Slows down critical conservation efforts and population monitoring

Our Goal:

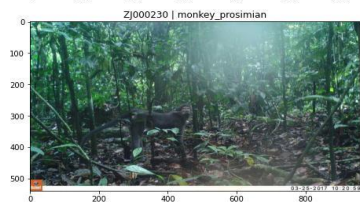
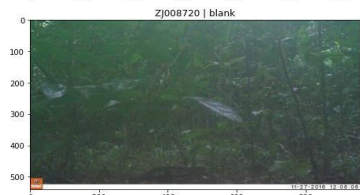
- Develop an automated image classification model to identify animal species in camera trap image

Target Classes (8):

- Antelope Duiker
- Bird
- Blank (No Animal)
- Civet Genet
- Hog
- Leopard
- Monkey Prosimian
- Rodent



Data Overview



Source: Camera trap images from **Tai National Park, Ivory Coast** sourced as part of DrivenData partnering with the Wild Chimpanzee Foundation and Max Planck Institute of Evolutionary Anthropology on Project Zamba, a multi-year effort to build machine learning tools for camera trap videos.

Dataset Size:

- **Total Available Images:** ~16,000
- Each image is accompanied by additional information:
- id - a unique identifier for each image
- filepath- image path including its split directory (train or test)
- site - the site in which the image was taken

Data Characteristics:

- Diverse angles, lighting conditions, and partial animal views inherent to camera trap imagery.
- Potential for class imbalance (some species rarer than others).

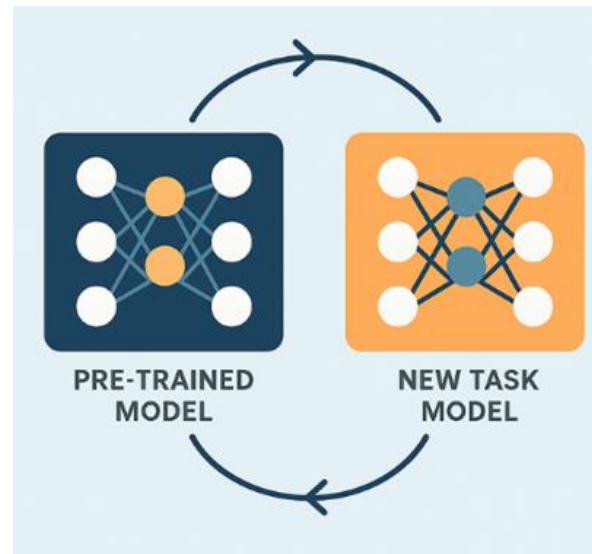
Project Methodology - Transfer Learning

Why Transfer Learning?

- **Limited Domain-Specific Data:** Training deep neural networks from scratch requires massive datasets (millions of images).
- **Leveraging Pre-trained Knowledge:** Transfer learning allows us to utilize powerful feature extraction capabilities learned by models on vast, general image datasets

Core Concept:

- Take 3 pre-trained Convolutional Neural Network (CNN) backbones
 - a. **ResNet50**
 - b. **EfficientNetV2S**
 - c. **VGG16**
- Freeze its initial layers to preserve general image features (edges, textures, shapes)
- Make multiple iterations to reduce validation loss
- Fine-tune parts of the frozen backbone with a very low learning rate to adapt it further



Data Exploration, Preparation & Splitting

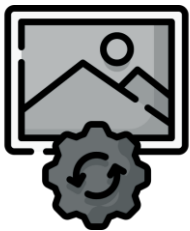
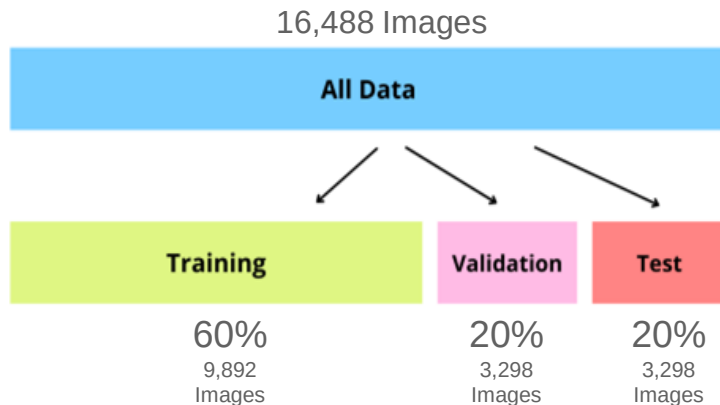


Image Preprocessing:

- **Resizing:** All images resized to 224x224 pixels (standard input for most CNNs).
- **Normalization:** Pixel values normalized using ImageNet's mean and standard deviation.

Dataset Splitting

- **Stratified Split:** Ensured that the proportion of each of the 8 classes was maintained across all splits to prevent bias.
- **Final Split Ratio (60/20/20):**
 - **Training Set (60%):** For model parameter learning.
 - **Validation Set (20%):** For hyperparameter tuning, early stopping, and monitoring generalization during training.
 - **Test Set (20%):** A completely unseen dataset for final, unbiased performance evaluation.



Animal Class	Training Count	Training %	Validation %	Test %
monkey_prosimian	2,492	15%	15%	15%
antelope_duiker	2,474	15%	15%	15%
civet_genet	2,423	14%	14%	14%
leopard	2,254	13%	13%	13%
blank	2,213	13%	13%	13%
rodent	2,013	12%	12%	12%
bird	1,641	9%	9%	9%
hog	978	5%	5%	5%

Models Explored - **VGG16** trained on ImageNet

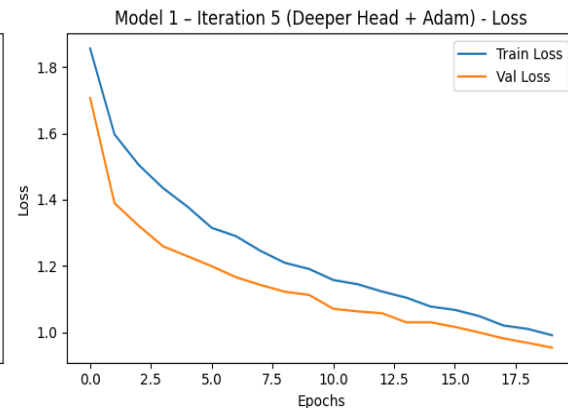
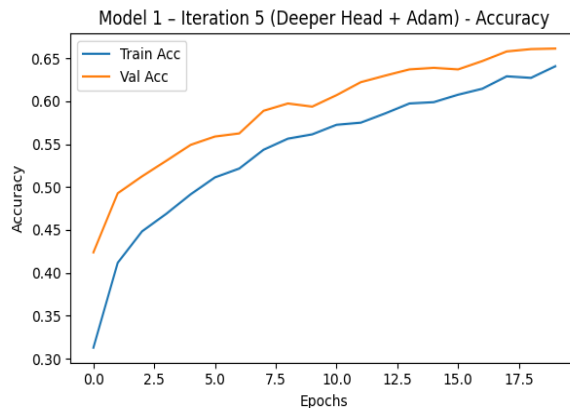
VGG16:

- **Architecture:** 16-layer CNN with stacked 3×3 conv layers and 2×2 max pooling. Simple and deep design
- **Output Features:** 4096
- **Strengths:** Strong feature extractor, easy to fine-tune, widely used for transfer learning.
- **Considerations:** High parameter count (~138M), computationally heavy vs. ResNet / EfficientNet.

Iteration A: Initial Baseline

- VGG16 base frozen, 20-epoch training with early stopping
- Custom head: GAP → Dense(256, ReLU) → BatchNorm → Dropout(0.3) → Dense(128, ReLU) → Dropout(0.3) → Softmax
- Deeper head with additional Dense layer to increase capacity
- Adam optimizer, learning rate $1e-4$

Metric	Value
Best Epoch	20
Training Accuracy	0.6407
Validation Accuracy	0.6613
Training Loss	0.9908
Validation Loss	0.9532

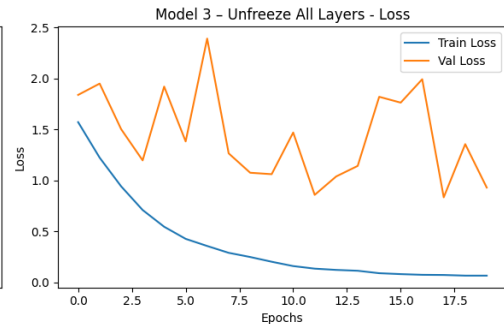
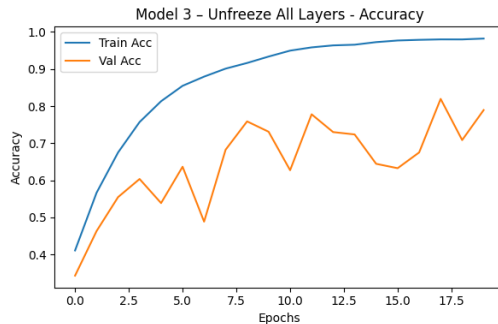


Models Explored - **VGG16** trained on ImageNet (Contd.)

Iteration B: Fine-Tuned VGG16 (fully unfrozen)

- VGG16 fully unfrozen, 20-epoch training with early stopping
- Custom head: GAP → Dense(256, ReLU) → BatchNorm → Dropout(0.3) → Softmax
- Regularized with BatchNorm + Dropout
- Adam optimizer, learning rate 1e-4

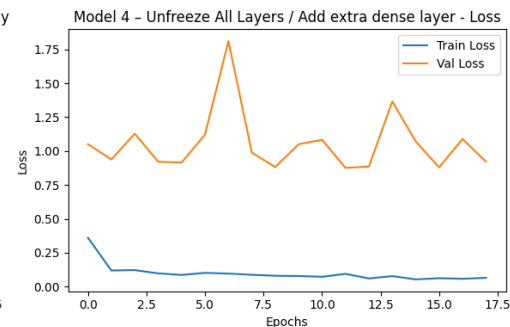
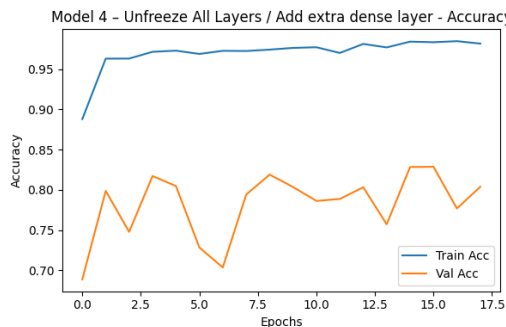
Metric	Value
Best Epoch	18
Training Accuracy	0.98
Validation Accuracy	0.8196
Training Loss	0.0725
Validation Loss	0.8342



Iteration C: Fine-Tuned VGG16 + Deeper head

- VGG16 fully unfrozen, 20-epoch training with early stopping
- Custom head: GAP → Dense(512) → BN → Dropout(0.5) → Dense(256) → BN → Dropout(0.5) → Softmax
- Added extra Dense layer to increase model capacity
- BatchNorm and higher Dropout used for regularization
- Adam optimizer, learning rate 1e-4

Metric	Value
Best Epoch	12
Training Accuracy	0.9703
Validation Accuracy	0.7887
Training Loss	0.0946
Validation Loss	0.8759

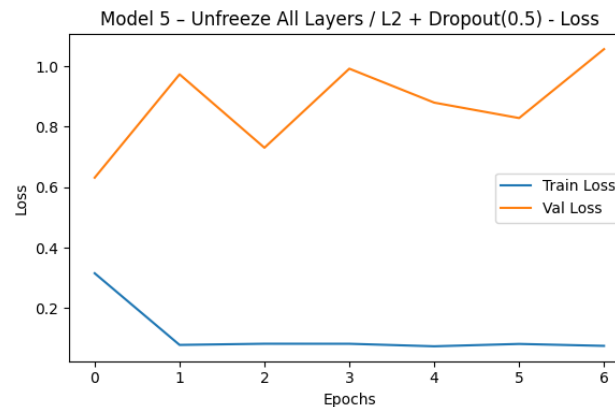
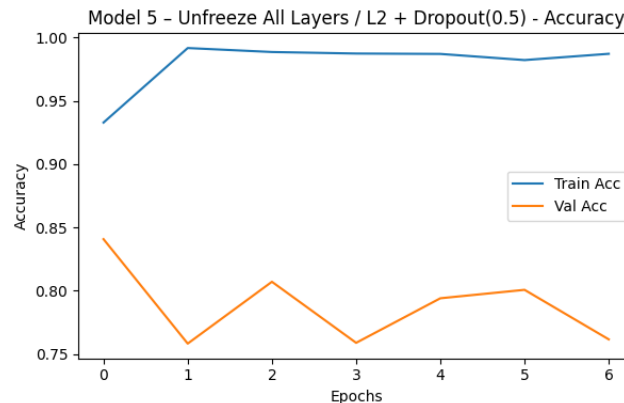


Models Explored - **VGG16** trained on **ImageNet** (Contd.)

Iteration D: Fine-Tuned VGG16 + L2 Regularization (Best Performer)

- VGG16 fully unfrozen, 20-epoch training with early stopping
- Custom head: GAP → Dense(256, L2) → BN → Dropout(0.5) → Dense(128, L2) → Softmax
- L2 regularization added to Dense layers to reduce overfitting
- Balanced architecture depth with smaller second Dense layer
- Adam optimizer, learning rate 1e-4

Model Iteration	Training Accuracy	Training Loss	Validation Accuracy	Best Validation Loss
A	0.6407	0.9908	0.6613	0.9532
B	0.98	0.0725	0.8196	0.8342
C	0.9703	0.0946	0.7887	0.8759
D - Best Performer	0.9329	0.3156	0.8408	0.6316

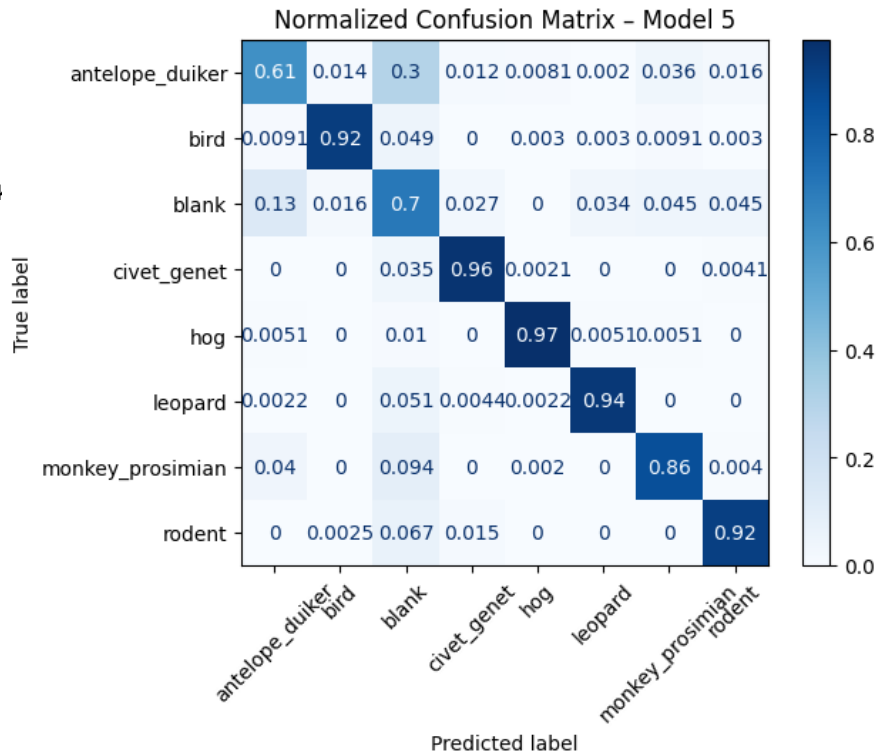


Models Explored - **VGG16** trained on ImageNet (Contd.)

Performance on Test Dataset (unseen 20% of dataset)

- Test Accuracy: **0.8472**
- High accuracy for **bird**, **civet_genet**, **hog**, and **rodent** ($\geq 90\%$)
- **Antelope_duiker** and **blank** show major confusion, especially with each other
- **Monkey_prosimian** and **leopard** moderately under-predicted, often confused with **blank**
- **Blank** is over-predicted, absorbing errors from several classes

Label	True Count	Predicted Count
antelope_duiker	495	302
bird	329	303
blank	441	309
civet_genet	485	466
hog	196	190
leopard	450	423
monkey_prosimian	498	428
rodent	402	370



Models Explored - ResNet50

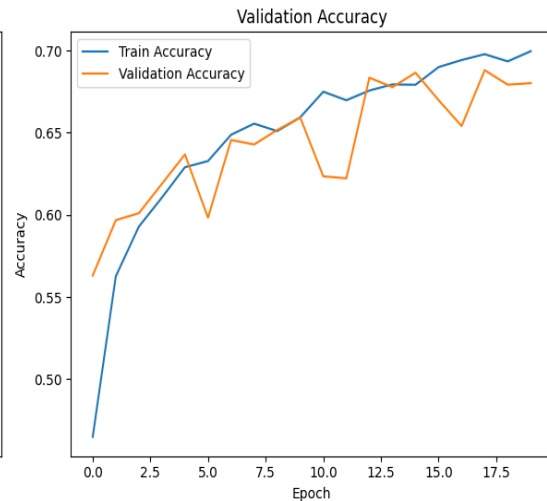
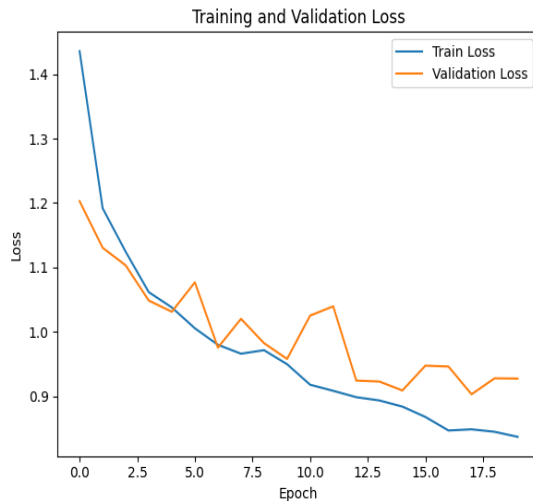
ResNet50:

- **Architecture:** 50-layer Residual Network. Known for skip connections to mitigate vanishing gradients.
- **Output Features:** 2048
- **Strengths:** Robust, widely used, good balance of depth and efficiency.

Iteration 1: Initial Baseline

- ResNet50, 20 epoch training with early stopping
- Unfrozen backbone, default head.
- No Batch normalization
- Adam Optimizer
- Validation Loss: **0.9030**
- Validation Accuracy: **0.6880**

```
--- Model 1 - Results Summary --- Baseline Model
| Metric | Value |
|-----|-----|
| Epoch of Best Val Loss | 18 |
| Validation Loss (min) | 0.903 |
| Validation Accuracy (at min val loss) | 0.688 |
| Training Loss (at min val loss) | 0.8486 |
| Training Accuracy (at min val loss) | 0.6977 |
```

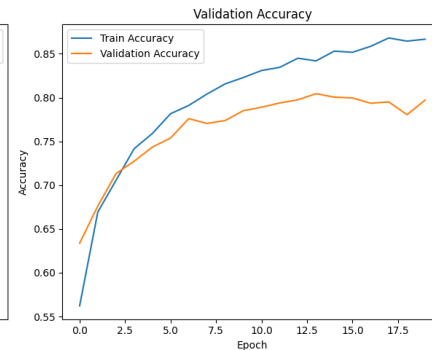
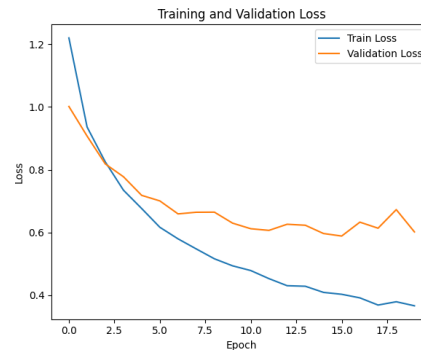


Models Explored - **ResNet50** (contd.)

Iteration 2: Optimized Frozen ResNet50

- ResNet50, frozen backbone.
- Custom head (100 neurons, dropout)
- ReLu Activation Function.
- Adam optimizer, LR 0.001.
- Validation Loss reduced to : **0.5888**
- Validation Accuracy improved to: 0.7996

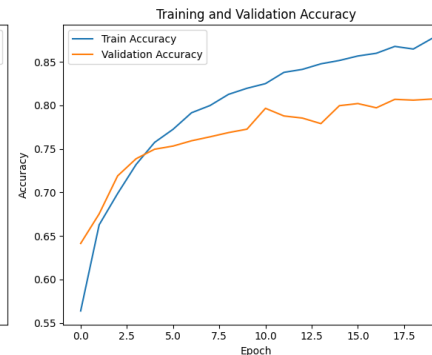
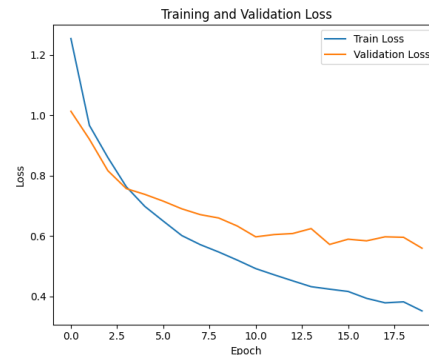
--- Model 2 - Results Summary	
Metric	Value
Epoch of Best Val Loss	16
Validation Loss (min)	0.5888
Validation Accuracy (at min val loss)	0.7996
Training Loss (at min val loss)	0.4033
Training Accuracy (at min val loss)	0.8517



Iteration 3: Slower Learning Rate

- ResNet50, frozen backbone.
- Custom head (100 neurons, dropout)
- ReLu Activation Function.
- Adam optimizer, **LR reduced to 0.0005.**
- Validation Loss reduced to : **0.5600**
- Validation Accuracy improved to: 0.8072

--- Model 3 - Results Summary	
Metric	Value
Epoch of Best Val Loss	20
Validation Loss (min)	0.56
Validation Accuracy (at min val loss)	0.8072
Training Loss (at min val loss)	0.3523
Training Accuracy (at min val loss)	0.8767

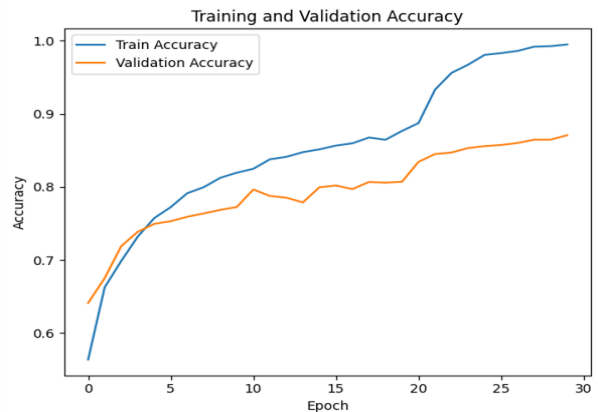
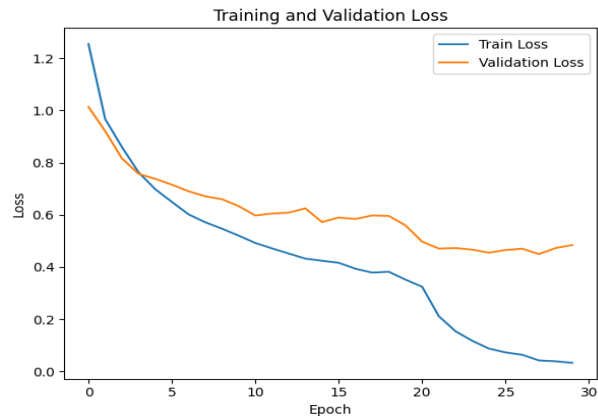


Models Explored - **ResNet50** (contd.)

Iteration 4: Fine Tuned (Best Performer)

- Unfreeze all layers in the backbone
- Very low LR (0.00001) for Fine Tuning
- Ran for additional 10 epochs with early stopping on Iteration 3 model
- Validation Loss reduced to :**0.4500**
- Validation Accuracy improved to: **0.8648**

Model Iteration	Training Accuracy	Training Loss	Validation Accuracy	Best Validation Loss
1	0.6977	0.8486	0.6880	0.6977
2	0.8517	0.4033	0.7996	0.5888
3	0.8767	0.3523	0.8072	0.5600
3 - Fine Tuned	0.9920	0.0429	0.8648	0.4500

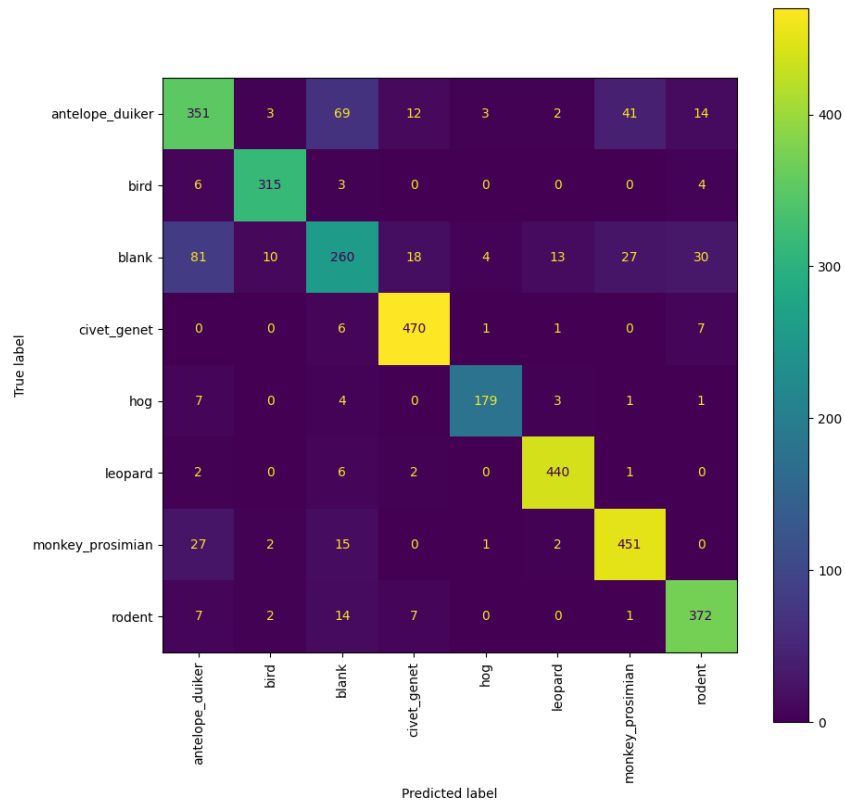


Models Explored - ResNet50 (contd.)

Performance on Test Dataset

- Tested on dedicated 20% dataset
- Test Accuracy: 0.8605
- Model is highly accurate for **bird**, **civet_genet**, **leopard**, and **rodent**
- **antelope_duiker** and **blank**: These two classes still show the highest mutual confusion, and also confusion with **monkey_prosimian**

Labels	Predicted Labels	True Labels
Antelope Duiker	481	495
Bird	332	328
Blank (No Animal)	377	443
Civet Genet	509	485
Hog	188	195
Leopard	461	451
Monkey Prosimian	522	498
Rodent	428	403



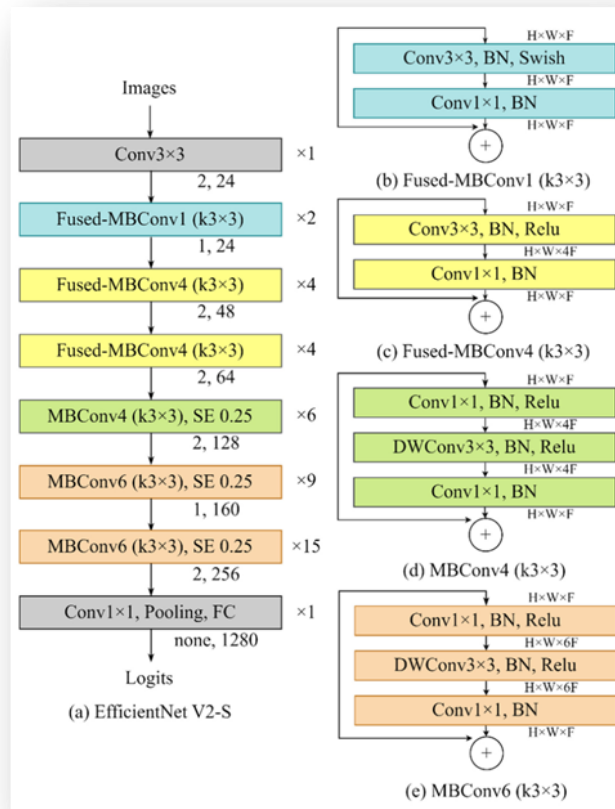
Models Explored - EfficientNetV2S (Overview)

EfficientNetV2s:

- **Model Origin:** Developed by Google Research, introduced in the paper
- EfficientNetV2: Smaller Models and Faster Training (ICML 2021)

Architecture Highlights

- 20.5 million parameters
- 385 Layers in total
- Composed of 508 operations across convolutional layers
- Uses MBConv blocks with squeeze-and-excitation and residual connections
- Input resolution: 384×384 pixels

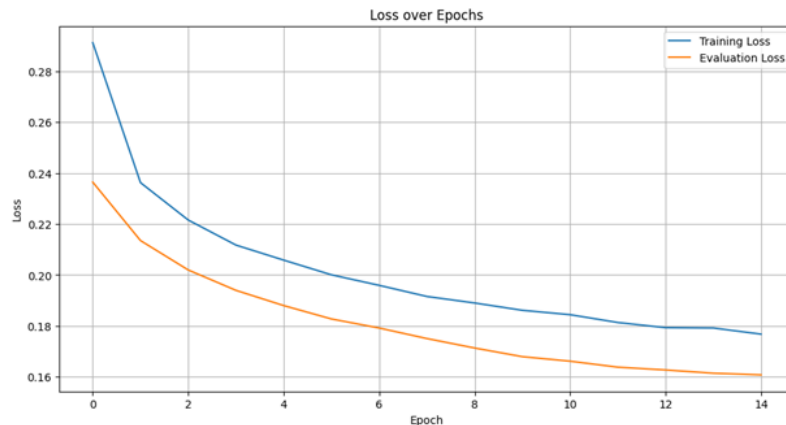
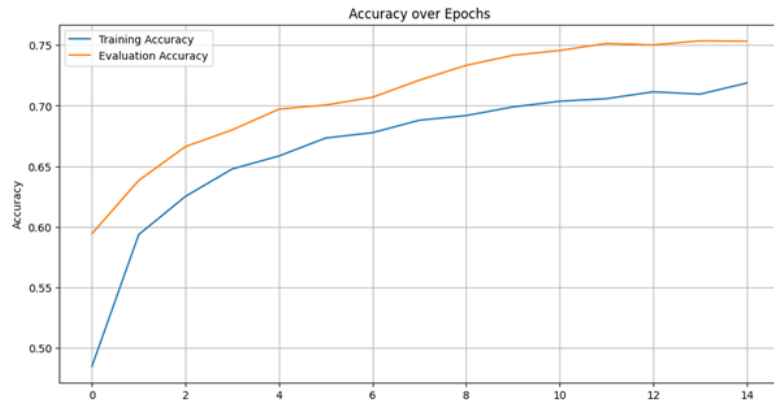


Models Explored - EfficientNetV2S

EfficientNetV2S Iteration 1:

- **Replaced input layer:**
 - 224 x 224
- **Replaced the head with the following**
 - Estimator: Adam
 - Global Average Pooling Layer 2D
 - Dense layer with 8 neurons (matching our number of labels)
 - Activation Function: Sigmoid
- **Frozen all layers of original model**
 - Trainable Parameters: 10 248
 - Non trainable parameters: 20 331 360
- **Number of epochs: 15**

Output Summary	Loss	Accuracy
Training	0.1762	0.72
Validation	0.1608	0.76

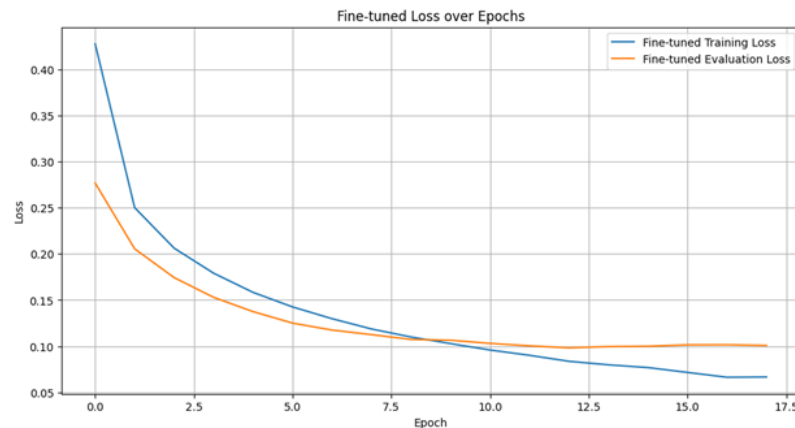
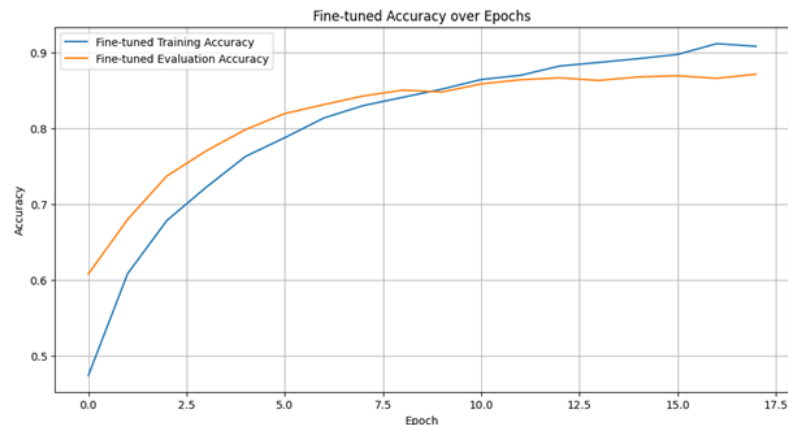


Models Explored - EfficientNetV2S

EfficientNetV2S Iteration 2 (fine tuned):

- Replaced input layer:
 - 224 x 224
- Replaced the head with the following
 - Estimator: Adam
 - Global Average Pooling Layer 2D
 - Dense layer with 8 neurons (matching our number of labels)
 - Activation Function: Sigmoid
 - Added Drop Out at 20%
 - Added Early stopping
- Unfrozen last 20 layers of original model
 - Trainable Parameters: 1 345 608
 - Non trainable parameters: 18 996 000
- Reduced learning rate
- Number of epochs: 30
 - Best result: After 18 Epochs

Output Summary	Loss	Accuracy
Training	0.0664	0.91
Validation	0.1007	0.87



Models Explored - EfficientNetV2S (Results Summary)

Model Iteration	Retrained Parameters	Best Validation Accuracy	Best Validation Loss
1	10 000	0.76	0.1608
2	1.4 million	0.87	0.1007

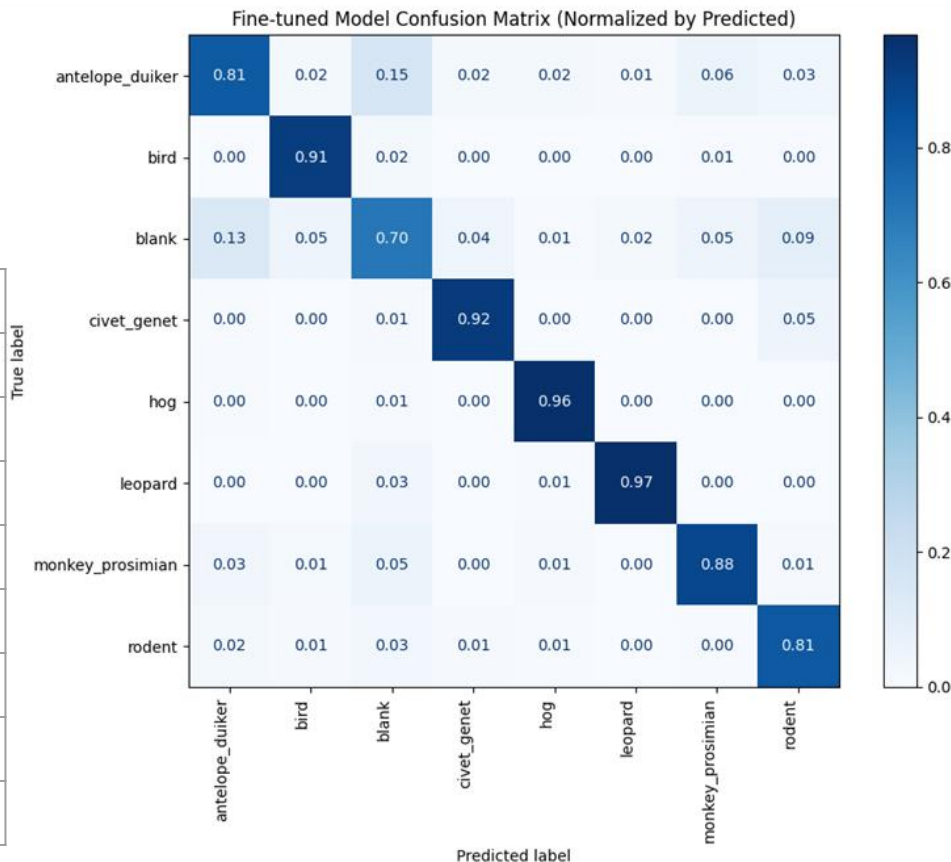
- **Conclusion:**
 - Unfreezing layers, adding dropout, and increasing epochs with early stopping improved the results.
 - With more training time, the following can also be tried to improve results further:
 - Unfreeze the entire model
 - Do incremental fine tuning of the already trained model

Models Explored - EfficientNetV2S

Performance on Test Dataset

- Tested on dedicated 20% dataset
- Test Accuracy: 0.87
- Model predicts well overall
- Not great with blank images, rodent and duiker

Label	True Count	Predicted Count
antelope_duiker	495	448
bird	329	333
blank	441	377
civet_genet	485	496
hog	196	204
leopard	450	456
monkey_prosimian	498	532
rodent	402	450



Comparative Model Performance

- **EfficientNetV2S** achieved the best test score (0.8665) and validation accuracy (0.8717).
- **ResNet50** delivered strong results with a frozen backbone and fine-tuning.
- **VGG16** showed overfitting: high training accuracy, but lowest test and validation scores.
- **EfficientNetV2S** offered the best generalization overall.

Model Backbone	Parameters	Best Training Accuracy	Best Training Loss	Best Validation Accuracy	Best Validation Loss	Test Score
ResNet50	Frozen backbone, Custom head (100 neurons, dropout), ReLu Activation Function, Adam optimizer, LR 0.0005, Fine-tuned @ 0.00001	0.9920	0.0429	0.8648	0.4500	0.8605
EfficientNetV2S	Unfrozen last 20 layers, Custom head (GlobalMaxPooling2D, Dropout (20%), output layer), Sigmoid Activation Function, Adam optimizer, low learning rate, early stopping, increased epochs	0.9142	0.0658	0.8717	0.0981	0.8665
VGG16	Fine-tuned VGG16 (unfrozen), Custom head (256→128, L2, Dropout), ReLU activation, Adam optimizer, LR 0.0001	0.9329	0.3156	0.8408	0.6316	0.8472

Conclusion & Future Work

Conclusion:

- Successfully developed an image classification model for Tai National Park camera trap images using transfer learning.
- Achieved a significant improvement in accuracy from baseline by implementing a structured iterative approach, proper validation, and early stopping.
- The model could assist researchers in efficiently processing large volumes of camera trap data.

Future Work & Next Steps:

- **Head Architecture Tuning:** Further experiment with different hidden layer sizes, activation functions, and dropout rates in the classification head.
- **Fine-tuning the Backbone:** Gradually unfreeze and train later layers of the best performing backbone with very low learning rates to adapt features more specifically to the dataset.
- **Data Augmentation:** Explore more aggressive or domain-specific augmentation techniques.

References

- Notebook: “SCS_3546_Wildlife_Project_EfficientNetV2s”
 - Notebook: “SCS_3546_Wildlife_Project_RESNET”
 - Notebook: “SCS_3546_Wildlife_Project_VGG16”
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Thank You
